Quantum Physics, 2026 Fall, BME
Lecture 02 - Postulates of quantum mechanics

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1 Linear algebra: relations between Hermitian and unitary matrices

1. Both are normal matrices, and hence can be diagonalized, i.e., have a *spectral representation* (also known as *diagonal representation*).
2. Given a Hermitian matrix H, it holds that e^{iH} and e^{-iH} are unitary matrices.
3. For a unitary matrix U, it holds that $i\log(U)$ and $-i\log(U)$ are Hermitian.
4. Proof: via spectral decomposition.

2 Classical master equation

1. An exercise on classical master equations, to be covered in this week's exercise class, will illustrate how to solve a set of coupled ordinary first-order linear differential equations. Solving the Schrodinger equation, to be presented as Eq. (3) of these notes, is analogous to this.

3 Postulates of quantum mechanics

1. Here, we consider quantum mechanics as a *mathematical framework* of physical theories. Some call it the 'operating system (OS)', on which the physical theories run as 'applications'.
2. This mathematical framework is based primarily on linear algebra and probability theory.
3. This section is based on Nielsen-Chuang section 2.2.

3.1 Postulate 1: The state is a complex unit vector

1. Postulate 1: The state of a system at a given moment of time is described by a complex unit vector.

2. Example: consider an electron trapped in a potential well. It cannot move. However, its *spin* is a degree of freedom that is described by quantum mechanics. The state of its spin is described by a unit vector $|\psi\rangle \in \mathbb{C}^2$.

3. Example: a *qubit* or *quantum bit* or *two-level quantum system* is an abstract quantum systems with states described by unit vectors in $\mathbb{C}^2$.

   Denote canonical basis vectors of $\mathbb{C}^2$ as $|0\rangle = (1,0)^T$ and $|1\rangle = (0,1)^T$.

   Then, a state is denoted as $|\psi\rangle = a\,|0\rangle + b\,|1\rangle$, which is a 'valid quantum state' (i.e., a unit vector) if the *normalisation condition* holds: $|a|^2 + |b|^2 = 1$.

4. Example: certain electrons in a benzene molecule can 'hop' between the six carbon atoms of the ring. Forgetting about their spins, these electrons can be described by quantum states (unit vectors) in $\mathbb{C}^6$.

5. Terminology: A linear combination $\sum_i \alpha_i\,|\psi_i\rangle$ is a *superposition* of states $|\psi_i\rangle$ with *amplitudes* α_i.

3.2 Postulate 2: Time evolution is unitary

1. Postulate 2: The time evolution of a closed quantum system is described by a unitary transformation. That is, the state $|\psi(t_1)\rangle$ of a system at time t_1 is related to the state $|\psi(t_2)\rangle$ of the system at time t_2 by a unitary matrix $U(t_2,t_1)$:

$$|\psi(t_2)\rangle = U(t_2,t_1)\,|\psi(t_1)\rangle\,. \tag{1}$$

2. 'Closed system': does not interact with other systems. This is an idealization, but a useful one.

3. Examples: for a qubit, whose state space is $\mathbb{C}^2$, Pauli matrices I, X, Y, Z can describe time evolution as they are unitary.

   Another often used 2×2 unitary matrix is the *Hadamard matrix* or *Hadamard gate*:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}. \tag{2}$$

4. Postulate 2 of unitary time evolution implies that if the initial state is normalized, then the state at any later time is also normalized. This follows from the fact that a unitary matrix preserves the scalar product, see Eq. (23) of Lecture 01.

3.3 Postulate 2': Time evolution is governed by the Schrödinger equation

1. Similarly to how Newton's second law governs time evolution in classical mechanics, the Schrödinger equation governs time evolution in quantum mechanics.

2. Postulate 2': The time evolution of the state of a closed quantum system is described by the Schrodinger equation:

$$i\hbar \frac{d\,|\psi\rangle}{dt} = H\,|\psi\rangle\,. \tag{3}$$

   Here, $\hbar \approx 1.05 \times 10^{-34}\,\mathrm{J}\cdot\mathrm{s}$ is the reduced Planck constant, and H is a fixed Hermitian matrix called the *Hamiltonian* of the system. (Not to be confused with the Hadamard gate.)

3. Remark: a more practical unit for the reduced Planck constant is μeV/GHz: then, we have $\hbar \approx 0.66\,\mu\mathrm{eV/GHz}$.

4. How to figure out the Hamiltonian of a physical system? Not part of the 'mathematical framework'. We'll deal with this later.

5. Postulates 2 and 2' are seemingly different. Do they contradict each other? No. In fact, it is easy to check that the solution of the Schrodinger equation reads

$$|\psi(t_2)\rangle = \exp\left[\frac{-iH(t_2-t_1)}{\hbar}\right]|\psi(t_1)\rangle\,. \tag{4}$$

   Furthermore, the exponential factor is a unitary matrix (due to item 2 in Section 1), hence this solution of Postulate 2' is in accordance with Postulate 2.

6. Exercise: Consider a qubit and assume its Hamiltonian is $H = -\hbar\omega Z$, where $\frac{\omega}{2\pi} = 1\,\mathrm{GHz}$. Compute the time evolution operator $U(t,0)$, the time-evolving state for the initial state $|\psi(0)\rangle = (1,0)^T \equiv |0\rangle$ and for the initial state $|\psi(0)\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. Plot the time-evolving states.

7. Some open systems can still be described with the Schrodinger equation, albeit with a time-dependent Hamiltonian. Some open systems require other tools to be described.

4 Postulate 3: Measurement in quantum mechanics

1. So far: evolution of a closed quantum system. Here: observation, measurement.

2. Nielsen-Chuang postulates quantum measurement in a more general way than we do here. What we postulate here is a special case of their description, which they call *projective measurement*.

3. Here, the phrase *observable* will denote a Hermitian matrix. Later, we will clarify that such observables represent physical quantities, such as position, momentum, magnetic moment, etc. Hence a *measurement described by an observable* will be reinterpreted as a *measurement of a physical quantity*.

4. Postulate 3: A (projective) measurement is described by an *observable* M, a Hermitian operator on the state space of the system being observed.

   The observable has a spectral decomposition,

$$M = \sum_j \lambda_j P_j, \tag{5}$$

   where P_j is the projector onto the eigenspace of M with real eigenvalue λ_j.

   The possible outcomes of the measurement are the eigenvalues λ_j of the observable.

   Upon measuring state $|\psi\rangle$, the probability of getting outcome λ_j is

$$p_j \equiv p(\lambda_j) = \langle\psi|P_j|\psi\rangle\,. \tag{6}$$

   Given that outcome λ_j was obtained, the state of the quantum system immediately after the measurement is

$$|\psi_{\text{post}}\rangle = \frac{P_j\,|\psi\rangle}{\sqrt{p_j}}. \tag{7}$$

5. Remark: if the jth eigenvalue is nondegenerate, then $P_j = |\varphi_j\rangle\,\langle\varphi_j|$, and hence

$$p_j = |\langle\varphi_j|\psi\rangle|^2 \tag{8}$$

6. Exercise: (*measurement of a qubit in the computational basis*).

   Consider the observable $C = (1 - Z)/2$. The measurement of C is often called *measurement of the qubit in the computational basis.*

   Express the matrix form of C, its eigenvalues, and its eigenvectors.

   Compute the measurement outcome probabilities and the post-measurement states, if the pre-measurement state is
   (a) $|0\rangle = (1, 0)^T$,
   (b) $|1\rangle = (0, 1)^T$,
   (c) $|\psi\rangle = \frac{1}{\sqrt{2}}\,|0\rangle + \frac{1}{\sqrt{2}}\,|1\rangle$, and
   (d) $|\psi\rangle = \frac{1}{\sqrt{3}}\,|0\rangle + \sqrt{\frac{2}{3}}\,|1\rangle$.

7. Example (*measurement of X on a qubit*) Repeat (a,b,c,d) for the measurement of the observable X.

8. Claim: p_j-s indeed form a probability distribution, that is, $\sum_j p_j = 1$.

   Proof: Sum up Eq. (6) for j, use the fact that the scalar product is linear in its second vector, and then use that $\sum_j P_j = 1$.

9. What is the relation between Postulates 2 and 3? Can we derive Postulate 3 from Postulate 2? This is an important and interesting question we do not address in this course.

10. Terminology:

    Instead of *projective measurement*, the phrases *von Neumann measurement* and *strong measurement* are used often.

    Collapse of the wave function refers to the immediate change of the state to the post-measurement state as the result of the measurement.

5 Expectation value, standard deviation

1. Recall the average $E(X)$ and standard deviation $\Delta(X)$ of a real-valued discrete random variable X.

   Assume the random variable X can take n different real values, x_j where $1 \leq j \leq n$.

   Assume that the probability distribution of X is given by the probabilities p_j. It holds that $\sum_{j=1}^{n} p_j = 1$.

$$\text{expectation value: } E(X) = \sum_j p_j x_j \tag{9}$$

$$\text{standard deviation: } \Delta(X) = \sqrt{\sum_j p_j [x_j - E(X)]^2}. \tag{10}$$

2. Claim: the expectation value of an observable M at a given state $|\psi\rangle$ is

$$E(M, |\psi\rangle) = \langle\psi|M|\psi\rangle\,. \tag{11}$$

   Proof:

$$E(M, |\psi\rangle) = \sum_j \lambda_j p_j = \sum_j \lambda_j \langle\psi|P_j|\psi\rangle \tag{12}$$

$$= \langle\psi| \left(\sum_j \lambda_j P_j \right) |\psi\rangle = \langle\psi|M|\psi\rangle\,. \tag{13}$$

3. Notation: instead of $E(M, |\psi\rangle)$, we often write $\langle M \rangle_\psi$ or simply $\langle M \rangle$.

4. Exercise: Using the solution of the exercise of item 6 in Section 3.3, plot the time evolution of expectation values of X, Y, and Z.

5. Claim: the standard deviation of an observable M at a given state $|\psi\rangle$ is

$$\Delta(M, |\psi\rangle) = \sqrt{\langle M^2 \rangle - \langle M \rangle^2}. \tag{14}$$

6. Exercise: Assume we prepare a quantum system in an eigenstate $|\psi\rangle$ of the observable M, with corresponding eigenvalue λ. What is the expectation value of M in state $|\psi\rangle$, and what is its standard deviation?

7. Exercise: Compute the expectation values and standard deviations for all observable-state pairs in items 6 and 7 of Section 4.

6 Heisenberg inequality

1. Definition: The *commutator* of two observables (Hermitian operators) A and B is defined as

$$[A, B] = AB - BA. \tag{15}$$

2. Claim: (*Heisenberg inequality* or *Heisenberg uncertainty relation*) Consider a quantum system described by state vectors in $\mathbb{C}^n$. If A and B are observables ($n \times n$ Hermitian matrices), then the product of their standard deviations in state $|\psi\rangle$ are bounded from below by the following inequality:

$$\Delta(A, |\psi\rangle)\Delta(B, |\psi\rangle) \geq \frac{|\langle\psi|[A, B]|\psi\rangle|}{2}. \tag{16}$$

   Proof: see Nielsen-Chuang Box 2.4 on page 89.

3. Interpretation: if we have many copies of the same state ψ, and measure A on a subset of them and measure B on a diffent subset of them, then the standard deviations ΔA and ΔB of those measurements will satisfy the Heisenberg inequality.

4. Nielsen-Chuang calls this inequality the *Heisenberg uncertainty principle*. Since it is a result derived from Postulates 1 and 3, we do not call it a principle.

5. You have seen a version of the Heisenberg inequality, in the form of $\Delta x \Delta p \geq \hbar/2$, in Thornton-Rex section 5.6.

7 Observables and physical quantities: spin of an electron

1. In quantum mechanics, (some) physical quantities are represented by Hermitian operators, also called observables.

2. Example: the magnetic moment associated to the spin of an electron is a vector, $\vec{\mu} = (\mu_x, \mu_y, \mu_z)$. In quantum mechanics, this is represented by a constant times the vector of Pauli matrices:

$$\vec{\mu} = (\mu_x, \mu_y, \mu_z) = -\frac{1}{2} g \mu_B (\sigma_x, \sigma_y, \sigma_z). \tag{17}$$

   Here, $g \approx 2$ is the g-factor of electron in vacuum, and $\mu_B \approx 9.3 \times 10^{-24}\,\mathrm{J/T} \approx 58\,\mu\mathrm{eV/T}$ is the Bohr magneton.

   For example, the *x component of the magnetic moment* is

$$\mu_x = -\frac{1}{2} g \mu_B \sigma_x = -\frac{1}{2} g \mu_B \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \tag{18}$$

3. In this example, the Hamiltonian is the energy of the magnetic moment. In classical electromagnetism, the energy of a magnetic moment placed in a static, homogeneous magnetic field $\vec{B}$ is

$$E = -\vec{\mu} \cdot \vec{B}. \tag{19}$$

Hence, e.g., the quantum Hamiltonian of a magnetic moment in a z-directional magnetic field $\vec{B} = (0, 0, B)$ is:

$$H = -\mu_z B = \frac{1}{2} g \mu_B B \sigma_z. \tag{20}$$

4. Exercise: consider a confined electron in a z-directional magnetic field of 0.1 T. Prepare it in the initial state $|\psi_0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. Compute the time evolution of the state, and plot the expectation values of σ_x, σ_y, σ_z in the time-evolving state, as functions of time.